AP Computer Science A
Ticket Out the Door - Student Exemplar
Set 4: Mixed Data Types

Name _____ Period _____

Skill 4.1 Exercise 1	
Treat each box independently. Write the output, or identify the compile-time error and correct it.	
`int cnt = 27.2;` `System.out.println(cnt);`	Compile-time error: a double cannot be assigned directly to an int. Fix: int cnt = (int)27.2; Then the output is 27.
`int cnt = (int)27.9;` `System.out.println(cnt);`	27 Casting truncates the decimal part; it does not round.
`double d = 5;` `System.out.println(d);`	5.0 The int value is widened to a double.

`int cnt = 101;`
`double d = cnt;`
`System.out.println(d);`
	101.0
	The value of cnt is widened to a double.

`double d = -17.9;`
`int cnt = (int)d;`
`System.out.println(cnt);`
	-17
	Casting to int truncates toward zero.

Skill 4.2 Exercise 1	
Treat each box independently. Write the output, or identify the compile-time error and correct it.	

`int x = 10;`
`int y = 7;`
`System.out.println(y/x);`
	0
	Both operands are ints, so 7 / 10 uses integer division.

`int x = 10;`
`int y = 7;`
`System.out.println(x/y);`
	1
	Integer division discards the fractional part.

`double x = 10;`
`int y = 7;`
`System.out.println(y/x);`
	0.7
	Since x is a double, this uses floating-point division.

`int x = 10;`
`int y = 3;`
`double z = x/y;`
`System.out.println(z);`
	3.0
	10 / 3 is evaluated as int division first. The result 3 is then widened to 3.0.

`int x = 10;`
`int y = 3;`
`System.out.println((double)x/y);`
	3.3333333333333335
	Casting x before division preserves the fractional part.

`int x = 10;`
`int y = 3;`
`System.out.println((double)(x/y));`
	3.0
	The division happens before the cast. Casting the result cannot recover the fraction.

`double x = 10.5;`
`int y = 3;`
`System.out.println(x * y);`
	31.5
	Multiplying a double by an int produces a double.

`double z = 10.0 / 3;`
`int i = z;`
`System.out.println(i);`
	Compile-time error at int i = z;
	Fix: int i = (int)z;
	Then the output is 3.

`int p = 3;`
`double d = 10.3;`
`int j = (int)5.9;`
`System.out.println(p + p * (int)d - 3 * j);`
	18
	(int)d is 10 and j is 5.
	3 + 3 * 10 - 3 * 5 = 18.

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Skill 4.3 Exercise 1	
Treat each box independently. Write the exact output or name the runtime exception. For an exception, suggest a change that allows the division to run.	

`int x = 0;`
`int y = 0;`
`System.out.println(x/y);`
	Runtime exception: ArithmeticException.
	Integer division by zero is undefined. Change y to 1; then the output is 0.

`double x = 0;`
`double y = 0;`
`System.out.println(x/y);`
	NaN
	Floating-point zero divided by zero is not a number.

`int x = 100;`
`double y = 0;`
`System.out.println(x/y);`
	Infinity
	A positive number divided by double zero gives positive infinity.

`int x = -100;`
`double y = 0;`
`System.out.println(x/y);`
	-Infinity
	A negative number divided by double zero gives negative infinity.

Skill 4.4 Exercise 1
Write one Java statement declaring a double constant named SALES_TAX_RATE with the value 0.06.
final double SALES_TAX_RATE = 0.06;
Is each code segment legal? If illegal, identify the line that causes the error and explain why.

A.
`final double PI = 3.14;`
`int r = 5;`
`double c = 2 * PI * r;`
`PI = c;`
 	
B.
`final int MAX_STUDENTS;`
`MAX_STUDENTS = 30;`
 	
Explanation: A is illegal. PI = c; tries to reassign PI after it was initialized. B is legal. A final local variable may be declared first and initialized later, once.

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Optional AP FRQ-Style Practice: Class Snack Budget

Original, scaffolded practice for Skills 4.1, 4.2, and 4.4. Write Java statements; a complete class or method is not required.

A class has a snack budget, a number of students, and a price per snack. Assume budget and snackPrice are positive and students is greater than 0. Use these variables in your calculations; do not hard-code the sample results.

```
int budget = 25;
int students = 8;
double snackPrice = 1.75;
```

(a) Declare a double constant named TAX_RATE with the value 0.06. Then declare and initialize a double variable priceWithTax that represents the price of one snack including tax.

```
final double TAX_RATE = 0.06;
double priceWithTax = snackPrice * (1 + TAX_RATE);
```

(b) Declare and initialize a double variable budgetPerStudent that represents the budget divided equally among the students. Preserve the fractional part using a cast. Then print budgetPerStudent.

```
double budgetPerStudent = (double)budget / students;
System.out.println(budgetPerStudent); // 3.125
```

(c) Declare and initialize an int variable wholeSnacks that represents the number of whole snacks the class can buy at priceWithTax. Use a cast to discard the fractional part. Then print wholeSnacks.

```
int wholeSnacks = (int)(budget / priceWithTax);
System.out.println(wholeSnacks); // 13
```

(d) A student writes the following statement for part (b). What value is stored? Explain why it does not preserve the fractional part, and write a corrected statement.

```
double budgetPerStudent = (double)(budget / students);
```

It stores 3.0. The int division 25 / 8 produces 3 before the cast.
Corrected: double budgetPerStudent = (double)budget / students;

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